

Aakash Agarwal

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Kiel, Germany

Education

- Kiel University** Kiel, Germany
 - Master of Science - Quantitative Finance (1.3 GPA)* Oct 2023 – Jul 2026
 - Courses: Mathematical Finance, Computational Finance, Portfolio Analysis, Financial Risk Management, Time Series Analysis*
- Doon University** Dehradun, India
 - Integrated Masters Program - Economics (8.5 GPA)* Aug 2018 – Jun 2023
 - Courses: Microeconomics, Macroeconomics, Financial Economics, Advanced Econometrics, Statistical Methods, Operation Research*

Skills Summary

- Languages:** Python, R, MATLAB, SQL
- Frameworks:** Scikit, Numpy, Pandas, Statsmodels, Matplotlib, Seaborn, Scipy
- Tools:** MS Excel, SPSS, PyCharm, Git, Jupyter Notebook, MySQL, Atlas.ti
- Certificates:** Google Data Analytics, Financial Management Specialisation, Financial Engineering & Risk Management

Experience

- Solactive AG** (Frankfurt)
 - Quant Intern - Fixed Income QIS (Full-time)* Oct 2024 - Mar 2025
 - Developed Quantitative Indexing Strategies:** Designed rule-based indices (Emerging Markets, Corporate Credit, Transition Bonds) tracked by active funds; co-developed a BDC index launched as a NASDAQ-listed ETF.
 - Backtesting & Optimization:** Conducted historical simulations to evaluate index performance and robustness; implemented cap factor optimization frameworks to ensure benchmark consistency.
 - Automation, Business Development & Market Research:** Developed Python-based automation for performance reporting; leveraged Git for index model version control; researched global fixed-income trends for institutional objectives.
- Pradip & Company** (New Delhi)
 - Financial Analyst (Part-time)* Feb 2021 - Nov 2022
 - Business Valuation:** Performed based on the DCF model, multiples of publicly listed peers, and precedent deals.
 - Financial Forecasting:** Improved accuracy of the financial dashboard by 20% and reducing risk exposure by 30%.
 - Projection Reports:** Prepared Reports and assisted on ad-hoc researches, resulting in 15% operational cost reduction.

Projects

- Option Pricing Toolkit - Python Package (Derivatives Pricing, Numerical Methods):** Developed a Python package for option pricing using CRR, Black-Scholes, and Heston models. Implemented pricing via Monte Carlo simulation, numerical integration, finite difference schemes, and Laplace transform, with capabilities for analysis such as implied volatility and arbitrage opportunities. (July '24)
- Statistical Arbitrage with Cointegration (Quantitative Analysis, Trading Strategy):** Developed a project using cointegration to identify stock pairs with stable long-term relationships. Implemented minimum profit optimization for trading thresholds and automated real-time trading signals. (July '24)
- Calibration of the Heston Stochastic Volatility Model (Financial Modeling, Quantitative Analysis):** Calibrated Heston model parameters for S&P 500 options, aligning model prices with real market data. Validated results through detailed visualization. (May '24)
- Impact of Carbon Emission on the Cost of Debt (Data Analysis, Machine Learning):** (Thesis Project) Explored carbon risk's impact on Indian debt financing. Built ML model to forecast lender responses to carbon emissions. (June '23)

Publications

- Paper: Study of Machine Learning Algorithms for Stock Trading (Algorithms, Financial Forecasting):** Analyzed stock market forecasting using various ML and deep learning algorithms. (International Journal of Financial, Accounting, and Management, 2021)
- Paper: Assessment of Telecom Industry in India – From 2007 to 2020 (Telecom Industry, Economic Analysis, Policy Impact):** Analyzed the drastic changes in India's telecom sector over 13 years, highlighting debt issues, government regulations, and market consolidation. (Asian Journal of Economics, Finance and Management, 2021)

Honors & Awards

- Pandit Mahanand Dangwal Meritorious Award (2020)
- LAMP Fellowship by PRS India (2022)

Leadership & Activities

- Research & Volunteer Work – Department of Economics** Jul 2018 - Dec 2022
 - Assisted for the Annual Status of Education Report Survey.*
- Multidisciplinary Research Platform - Wiser World** Apr 2020 - Aug 2021
 - Built a research community, collaborated with 50+ interns.*